Diseases of geranium

1. Alternaria leaf spot - *Alternaria tenuis, A. alternata, Macrosporium pelargonii*

2. Pythium black leg - *Pythium irregulare*, *P. debaryanum*, *P. splendens, P. myriotylum*

3. Pythium root rot - *Pythium ultimatum*

4. Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

5. Rust - *Puccinia pelargonii-zonalis*

6. Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

7. Bacterial leaf spot - *Xanthomonas campestris* pv. *pelargonii*

or *Xanthomonas hortorum* pv. *pelargonii*

Alternaria leaf spot - *Alternaria tenuis, A.alternata, Macrosporium pelargonii*

Alternaria leaf spot - *Alternaria tenuis, A.alternata, Macrosporium pelargonii*

Symptoms

- The spots are more conspicuous on the lower leaves on the plants, but the upper young leaves may also be infected.
- Initially the infection appears as water- soaked area, less than 1mm in diameter on the lower surface of the leaf
- Spots enlarge upto 6-10mm in diameter and become irregular and the necrotic spots usually limited by the large vein of the leaf.

Alternaria leaf spot - *Alternaria tenuis, A.alternata, Macrosporium pelargonii*

Symptoms contd…

- Some of the spots have several concentric rings of darker brown tissues surrounding a dark center.
- Severe infection may kill the leaf or cause considerable leaf fall.
- Dead leaves shrivel and blacken and fungus sporulates profusely.

Alternaria leaf spot - *Alternaria tenuis, A.alternata, Macrosporium pelargonii*

Pathogens

Alternaria tenuis, A. alternata

- Mycelium is septate and pale to olivaceous brown in colour.
- Conidiophores arise singly or in small fascicles through stomata from the mycelium present in the dead centres of spots.
- Conidia are olivaceous brown, smooth having transverse and few longitudinal septa.

Alternaria leaf spot - *Alternaria tenuis, A.alternata, Macrosporium pelargonii*

Disease cycle and epidemiology

- The pathogen(s) overwinter in infected plant debris.
- Warm and humid climatic conditions favour the development of the disease.

Alternaria leaf spot - *Alternaria tenuis, A.alternata, Macrosporium pelargonii*

Management:

- Collect and destroy the infected plant debris.
- Apply mancozeb (0.25%) or captan (0.2%) or chlorothalonil (0.2%) as foliar sprays as the disease starts appearing.

Damping Off or Blackleg - *Pythium irregulare, P.debaryanum, P. splendens, P. myriotylum*

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Damping Off or Blackleg - *Pythium irregulare, P.debaryanum, P. splendens, P. myriotylum*

Geranium - Black leg (Pythium)

Black leg – geranium cuttings caused by Pythium infection

Damping Off or Blackleg - *Pythium irregulare, P.debaryanum, P. splendens, P. myriotylum*

Symptoms

- Root rot causes yellowing, stunting, wilting of above ground portions of plants.
- Plants have brown to black lesions at the base.
- A off white web growth can be seen infected plants.
- Leaves turn brown and rot.
- Roots have water soaked distinct lesions
- Roots turn brown and shrivel followed by decay

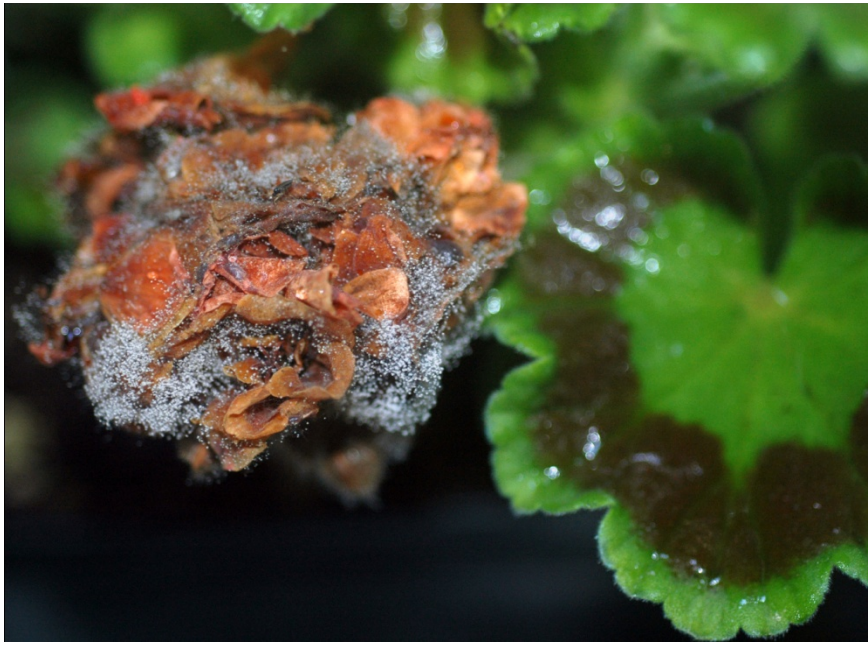
Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Botrytis blight on geranium leaf

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Gray, fuzzy growth of the fungus along this petiole.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)



Mature flowers are more likely to show symptoms than immature flowers.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Spent flowers colonized by Botrytis can be a source of infection for leaves.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Symptoms:

- Symptoms can develop on any plant part at any stage

Leaves

- Symptoms vary from discrete spots or lesions to large, dead areas, often with concentric rings
- V-shaped lesions can also develop and can be **confused with Bacterial Blight**
- Lesions often occur when spent flower petals fall onto the leaves, frequently the entire leaf and petiole get infection

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Symptoms on Flowers:

- Usually first evident **as premature fading and drying of flowers**
- Flowers turn brown and drop prematurely
- During periods of high moisture and relative humidity, senescing flowers are covered with a gray, fuzzy mass
- Infected petals wilt and drop on the foliage where they serve as a source of infection to leaves.
- The early fading or drying up of blossom is a common feature.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Symptoms on cuttings and Stems

- Symptoms appear at the base of **cuttings as light to dark brown lesions** which can result in complete basal rot
- **Stubs on stock plants** can develop brown lesions after cuttings are taken
- Under conditions of high moisture and humidity, any infected plant tissues and debris can develop the diagnostic fuzzy gray growth characteristic of this fungus.
- The fungus grows and sporulates rapidly on old and dying leaves, stems and dying floral parts.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Pathogen:

- *Botrytis cinerea* Pers. ex. Fr. (perfect stage: *Botryotinia pelargonii* Roed).
- Conidia are aseptate, hyaline mostly 8-14 x 6-9 µm in size.
- The fungus also forms sclerotia just beneath the cuticle or epidermis.
- The fungus survives either through the mycelium or sclerotia.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Disease cycle and epidemiology

- The disease is more apparent **during propagation and flowering.**
- High moisture level in greenhouse predisposes the plants to infection.
- Wounds of any kind predispose tissues to infection and colonization by the pathogen.
- Pathogen may also become saprophytic but can **seriously damage** the healthy geraniums, particularly those **tissues that are soft and succulent.**
- The sporulation of the pathogen is maximum at 20-25°C but decreases with high temperature 30°C.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Management

- Regulate the relative humidity in greenhouse that can be accomplished by proper spacing and watering and by ventilation and heating
- It is also essential to evacuate the warm, moist air in the evening, replacing it with cool, dry air that is then heated to a desirable temperature.
- Proper sanitation
- Grow tolerant disease varieties of geranium like Hawkeye, Waltztime, Hazel, Marian and Pearl Bailey.

Botrytis blight (Stem, Leaf, Flower) - *Botrytis cinerea* (PS: *Botryotinia pelargonii*)

Management contd…

- Fungicides such as Benomyl (0.05%), chlorothalonil (0.2%) or their combination can be used as protectants against infection.
- Captan (0.25%) with certain copper and sulphur mixtures (0.2%) and Vinclozolin (0.2%) alone is efficacious in protecting the plants.

Rust - *Puccinia pelargonii-zonalis*

Rust - *Puccinia pelargonii-zonalis*

igure 2. Rust spots (lesions) on the upper leaf surface of a geranium (courtesy Dr. F.F. Laemmlen)

Rust - *Puccinia pelargonii-zonalis*

Puccinia pelargonii-zonalis - Wikipedia

Rust - *Puccinia pelargonii-zonalis*

Rust - *Puccinia pelargonii-zonalis*

- The disease was first discovered in South Africa and later spread to different parts of the World
- It is a serious problem because of disfiguring of foliage by rust pustules
- In addition **many states have quarantine** to prevent the free movement of material by the growers till it's eradication

Rust - *Puccinia pelargonii-zonalis*

Symptoms

- The typical symptoms of the disease are white flecks, pinhead size usually on the upper surface of leaves
- Small, white or chlorotic blister-like pustules on the lower surface of the leaves
- As the chlorotic spots enlarge, the pustules in the centre enlarge in size upto 5-8mm, erupt and become reddish brown with spores.
- Uredia are then observed in a more or less circular fashion around the original sorus.

Rust - *Puccinia pelargonii-zonalis*

Symptoms contd..

- Most rust spots are formed on the lower surface of the leaves although they occasionally form on the upper surface.
- Heavily infected leaves turn chlorotic drop prematurely and leaving the plants partly defoliated.
- The spots are more apparent on leaves but also may develop on petioles, stipules and stems.

Rust - *Puccinia pelargonii-zonalis*

Pathogens

- *Puccinia pelargonii-zonalis* Doidge.
- The fungus produces subglobose to ovate, light brown uredospores measuring 24.5x22.0 µm, thin echinulated with two equatorial germ pores.
- **Telial stage is comparatively very rare**.
- The fungus is **microcyclic**, teliospores are 36-50 x 16-24 µm in size

Rust - *Puccinia pelargonii - zonalis*

Disease cycle and epidemiology

- Heavy spore production and pustule formation occurred within 14 days of infection at 21°C.
- The germination of these spores are accelerated in water at an optimum temperature of 16°C although the infection is more at 21°C than the higher temperature of 27 ^{o}C.
- The fungus forms appressoria and penetrated through the stomata by intercellular hyphae and intracellular haustoria.
- **The fungus overwinters in diseased leaves in form of uredospores.**

Rust - *Puccinia pelargonii-zonalis*

Management:

- Use resistant cultivars for planting such as Crimson Fire, Yours Truly, Sincerity and Springtime Irene.
- Follow thermotherapy with water saturated air at 38°C / 24 hr or dip in hot water at 50°C/90 sec of course there is **little injury to small expanding leaves with the hot water treatment but no effect on plant growth retardation is observed**.
- Application of mancozeb (0.25%), tridimefon, oxycarboxin and triforine (0.05% each) fungicides at two weeks intervals give preventive or curative effect.
- Bitertanol (0.05%) can be applied as spray at 30 days intervals.

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Symptoms

- The most common symptoms are bright yellow spottings on leaves, severe necrosis of shoot terminals or else extreme dwarfing of plants accomplished by foliar discoloration or wilting.
- The collapse of few leaves in the middle or upper portion of the main stem or side branches of the plant is noticed in some cultivars.
- At first, the petiole remain normal but after few days, they turn yellow and wilt followed by a gradual drying of the infected leaves and petioles resulting in leaf drop and defoliation of the affected area of the plant.

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Symptoms contd..

- In final stages the affected plants may show the dieback symptoms or drying tips of branches and brown to black discolouration of the main stem and branches.

- The browning is limited to the vascular tissues.

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Pathogen

- *Verticillium albo-atrum, V. dahliae*
- Conidiophores, branches and phialides are commonly in verticals.
- Conidia are aseptate, hyaline, in slimy masses. *V. albo–atrum* does not form microsclerotia while *V. dahliae* forms true **microsclerotia.**

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Disease cycle and epidemiology

- The pathogen(s) overwinter in the soil.
- Summer months are more conducive for the spread of the disease.

Verticillium wilt - *Verticillium albo-atrum, V. dahliae*

Management

- Use culture-indexed plants as stock material or certified disease free seed, sterile soil or planting media and strict sanitation measures.
- Suppress the symptoms by drenching with Benomyl at 1.5 g a.i/4.5l at the rate of 200ml/plant.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Fig. 1. Water-soaked spots on the underside of a *Pelargonium* leaf associated with the early-stage infection with *Xanthomonas campestris* pv. *pelargonii*.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

V-shaped yellowing that ends on the vein.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Fig. 3. Characteristic wilting in a single leaf (A) of a geranium in the greenhouse and in the whole plant (B), associated with *Xanthomonas campestris* pv. *pelargonii* infection. In early stages, affected plants may recover from wilting in the cool of the evening.

Fig. 4. Black stem rot associated with the advanced stages of *Xanthomonas campestris* pv. *pelargonii*.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Internal vascular discoloration of florist's geranium.

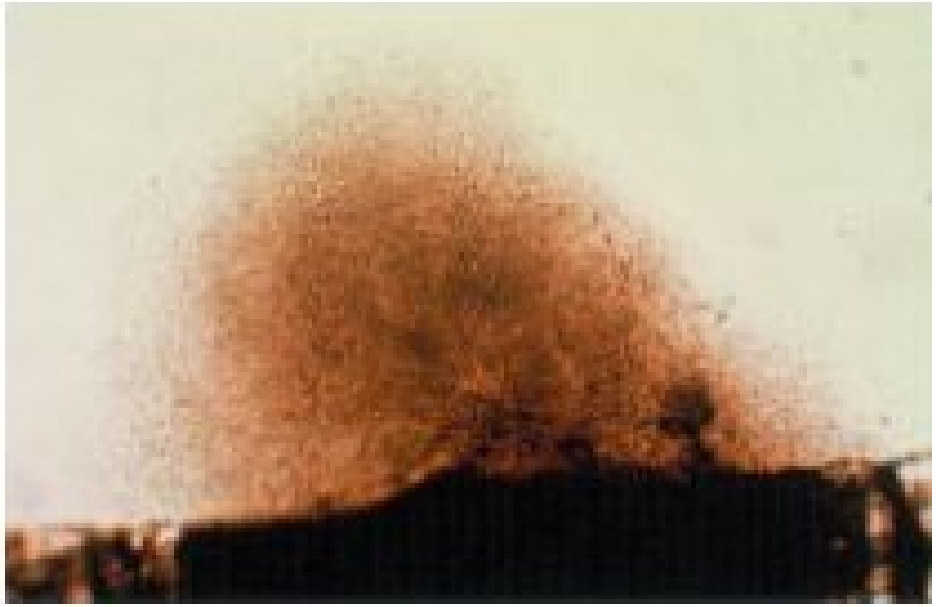


Fig. 5. Bacterial streaming from the cross section of a leaf spot from a bacterial blight symptomatic *Pelargonium*.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Symptoms

- Leaves, stem and cuttings are susceptible
- The infected leaves either produce round leaf spots or large, angular dead area depending upon the variety.
- The spots appear as small water soaked on undersides of the leaf having well defined margins, slightly sunken with 2- 3mm in dia. and rarely coalesce together.
- Later it is followed by necrosis and wilting of the leaves, which abscises or droops and hangs on the plant for a week or two.
- The original spots turn brown to black and become hard and dry.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Symptoms contd…

- Stem rot symptoms has given the term black rot to the disease.
- The stem rapidly blackens and shrivels into a dry rot, leaving the fibers and epidermis intact and the rest as a disorganized non-viscous mass.
- Thereafter, pith and cortex turn black and the exterior portions of the stem become grey and dull.
- Vascular bundles in stems and branches of plants infected with the bacteria become brown to black within 2-4 weeks of infection.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Symptoms contd….

- **The roots become are blackened but not decayed and the plants are often difficult to uproot.**
- Symptoms if present on cuttings fail to root and slowly rot from the base upward.
- The leaves wilt and often show the typical angular necrosis and wilt.
- Stem become dull black-brown just like the stem rot phase.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Pathogen

- Bacteria can grow best at 25-27°C.
- Bacterium is an aerobic, gram-negative, non-spore forming round rods, 1-1.5 x 0.5 - 0.7 µm in size, single or in pairs or sometimes in short chains, motile with polar flagellum.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Disease cycle and epidemiology

- **Seed transmission** is also an important mode of spread of the pathogen.
- The dissemination of the pathogen is carried out by splashing water either during propagation in the misted area or by routine overhead watering procedures.
- The symptoms are more evident in cool conditions on plants and if exposed to warm conditions the disease resulted in sudden wilting and death of the plants.

Bacterial Blight of Geranium (Bacterial Leaf Spot, Stem Rot or Wilt) *Xanthomonas campestris* pv. *pelargonii* or *Xanthomonas hortorum* pv. *pelargonii*

Management:

- Removal of all infected stock plants and debris, scrubbing down benches, clean up clay and plastic pots by steaming or sodium hypochlorite solution or formaldehyde soak for 10 minutes is essential.
- Provide good ventilation in greenhouse with good cultural practices.
- Use healthy and **indexed cuttings**
- Avoid over crowding.
- Disinfect the cutting knives or other instruments by dipping in 70% alcohol and flaming